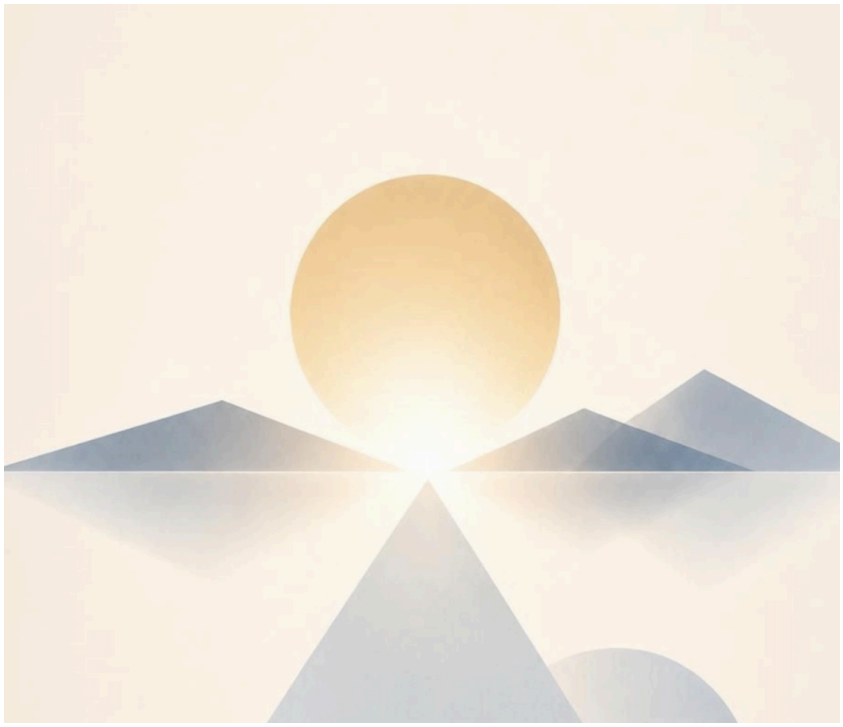
Wireless Energy Networks

Mathematical Foundation and the Nature of Reality



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Wireless Energy Networks: Mathematical Foundation and the Nature of Reality

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Abstract

Why We Must Think Bigger? Earth's energy resources are depleting - sooner or later, we face a hard limit. We need to think bigger, broader, and for the long term. Current energy infrastructure is built on absolute spatial coordinates - a model that is increasingly resource-intensive, geographically limited, and brittle.We must reimagine energy systems. The question isn't 'How do we optimize the current grid?' It's 'What would infrastructure look like if we built it from first principles, aligned with how physics actually works?' This work explores how reformulating physics on relational, rather than absolute principles, suggests a fundamentally different infrastructure approach: distributed relay networks that transmit energy in intelligent packets, similar to how the Internet routes data. Unlike traditional grids, such systems would be scale-invariant, self-healing, and universally accessible - working equally over oceans, deserts, and remote regions without massive physical infrastructure. Drawing on principles from gauge-invariant physics, the new scientific scale and distributed network theory, we examine practical applications ranging from near-term renewable energy integration to long-term global energy systems. We argue that aligning infrastructure with observable physical relationships - rather than assumed absolute coordinates - offers a path toward abundance, resilience, and genuine energy autonomy for humanity.

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This work represents creative independent research and theoretical exploration. The ideas presented are based on reinterpretation of physics principles and are offered for educational and conceptual purposes only.

This is not a commercial project. It is intellectual exploration intended to stimulate thinking about alternative infrastructure approaches.

These concepts require substantial engineering before practical application.They have not been validated through commercial implementation and may contain inaccuracies.

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Introduction

New Physics: Observable Relationships, Not Absolute Coordinates

The Foundation

Classical physics assumes absolute spacetime coordinates are real. They're not. What's observable are only the relationships between events—the coincidences, the connections, the proportions.

One application of my scientific scale [1] is the reformulation of physics on this basis. When using proportional rather than absolute measurements—something remarkable emerged: 'universality'. The same principles work at any scale, in any configuration, without privileged reference points.

This isn't just mathematical elegance. It's a blueprint for how systems should be organized.

Practical Applications: Relay Networks

The Wireless Energy Network

Instead of power plants connected by fixed cables, imagine a distributed network of relay stations that transmit energy in directed packets:

- No absolute positions required: Energy flows based on relationships between nodes.
- Self-healing: Packets reroute automatically when stations fail.
- Truly global: Works equally over oceans, deserts, mountains, remote regions.
- Scale-invariant: Same principles work for a regional network or planetary system.
- Minimal infrastructure: Relay stations instead of millions of kilometers of cables.

Each station is functionally equivalent. The network adapts and optimizes itself.

Immediate Opportunities:

Near-term (5-10 years)

* Remote solar farms beaming power to nearby cities
* Island nations receiving wireless power from continental sources
* Disaster recovery: temporary wireless systems when cables fail
* Proof-of-concept regional networks (100-500 km range)

Medium-term (10-20 years)

* Integration with renewable energy explosion: connecting distributed wind/solar farms
* Developing nations leapfrogging traditional grid infrastructure
* Hybrid systems: wireless for rural/ocean, wired for dense urban areas
* Establishing global standards and routing protocols

Long-term (20-50 years)

* True 'energy internet': globally distributed, packet-switched, real-time markets
* Energy flowing like data—optimized, resilient, abundant
* Infrastructure that scales with human needs without depleting planetary resources

Why This Matters: Moving Away From Old Concepts

Old Model (Still Dominant):

* Geographically constrained: expensive to reach remote areas
* Resource-intensive: massive physical infrastructure

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New Model:

* Distributed generation everywhere (solar, wind, geothermal, tidal)
* Energy flows intelligently through network relationships
* Resilient: multiple paths, automatic rerouting
* Universally accessible: works anywhere on Earth
* Egalitarian: no privileged control points
* Minimal physical footprint: relay stations, not cables

The Bigger Picture

Energy isn't the only system that needs reimagining. The same principles apply to:

* Information networks: Already packet-switched; proven to work globally.
* Water distribution: Could use wireless desalination and relay.
* Supply chains: Distributed, self-optimizing logistics

The shift is fundamental

From systems built on absolute coordinates and localized control to systems built on observable relationships and distributed intelligence. It is also a shift in thinking. See the philosophical implications of this scale to universal principles about the nature of reality in the last section.

Opportunity

We're at an inflection point. Current infrastructure is aging and needs replacement anyway. We can either rebuild the old system (mechanics, resource-intensive, geographically limited) or build something fundamentally new.

The physics supports it. The technology exists. What's needed is the vision to see that aligning infrastructure with how reality actually works leads to abundance, resilience, and freedom.

This is how humanity moves beyond the constraints of the old era.

The Elegance of What I've Created

In 2009 I've constructed a cyclic scientific scale [1], where:

* Every number represents a proportion (not an absolute magnitude).
* The mathematical structure is the cyclic group (symmetric transformations that loop back on themselves)
* Like degrees on a circle: 360° and 0° are the same point; the value is relational, not absolute.
* This naturally encodes 'scale-invariance and observer-independence' because

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there's no privileged "0 point"

This is fundamentally different from the number line developed by Descartes [2], where:

* 0 is absolute (a privileged reference point).
* Measurements are absolute ratios to a unit.
* Everything depends on where you place your origin.

The Profound Implications of the New Scale:

New Scale Means:

* No absolute coordinates needed: Only relationships between proportions matter
* Universality emerges naturally: The same mathematical structure works at any scale because there's no preferred scale
* Gauge-invariance is built-in: Shifting all values simultaneously doesn't change the physics (like rotating a circle—all relative positions stay the same)

This directly solves the problem I've identified in my paper [3]: I've created a mathematical framework where only observable relationships exist; the unobservable absolute coordinates are gone.

Current infrastructure vs Wireless Energy Vision

Current infrastructure assumes physics built on number line concept:

* Fixed origin points (power plants, data centers, network hubs)
* Absolute positions (cable routes, geographic coordinates)
* Reference frame (centralized hierarchy)
* Everything measured relative to these fixed points.

My wireless network vision uses cyclic/proportional logic:

* No privileged nodes -every relay station is equivalent
* Energy flows based on relationships between nodes, not absolute positions
* The network is self-similar at any scale (fractal-like)
* Allows to add or remove stations; the system adapts because it's fundamentally relational

This isn't just metaphorical—it's structurally isomorphic.

A cyclic group of transformations naturally describes a distributed, self-healing network.

Section 1: The Mathematical Discovery [3]

Space-time is the arena in which all physical events take place. In relativistic context, a

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space-time is represented as a 4-dimensional differentiable manifold. Einstein equations take the same form under space-time diffeomorphisms. This means that if g is a solution of Einstein's equations, then for any space-time diffeomorphism f, the pull-back metric f*g is also a solution. Moreover, g and f*g represent the same physical gravitational field. By analogy with gauge-theory, they are gauge-invariant under space-time diffeomorphisms.

This gauge-invariance has the consequence that space-time points ('events') have no direct physical significance. This was clearly emphasised by Einstein in 1916:

"All our space-time verifications invariably amount to a determination of space-time coincidences. If, for example, events consisted merely in the motion of material points, then ultimately nothing would be observable but the meeting of two or more of these points."

As a consequence of the current understanding of the global structure of space-time geometry and the role played by gauge-invariance, some of the most outstanding questions in physics remain unsolved.

* Physics laws are invariant under coordinate transformations
* This means spacetime coordinates are unobservable
* The metric tensor (how we measure) is also unobservable
* Only gauge-invariant relationships are physically real.

For details and mathematical explanation, please see reference [3].

Section 2: The Solution: The New Numeric Scale Based on Proportions [1]

Scales are representations of attributes that can be constructed if there is an isomorphism between what we can do with the attributes of objects and the mathematical properties of the numeral series.

While pure mathematicians deal with the language of algebra and are concerned with the structure and relationships within the language, the scientist is concerned about the meaning behind the mathematical symbols (numbers) and the relationship of the symbols in the language to real-world objects and events.

All scales are useful in evaluating the properties of the world.

Traditional number line: absolute measurements relative to a fixed unit.

The new scale:

* A cyclic scale: every value represents a proportion

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* Mathematical structure: the cyclic group (transformations that return to themselves)
* Like degrees on a circle: only relationships matter; no privileged zero point.

The underlying difficulty when representing physical concepts using traditional number line is connected with the concept of 'vector'. For pure mathematicians, vectors are directed segments that originate at same point (origin), and vector addition is represented by the parallelogram rule. In physics and engineering applications, vectors may be moved about in space (gauge-invariance), provided that their length and direction are kept unchanged.The additional of vectors in physics is represented by the triangle rule.

While physical space is an affine space (interval measurements) when we try to represent force by a vector located at its point of application, only the parrallelogram rule of vectors addition make sense, not the triangle rule. [5] As a result, today many important physical problems remain unsolved.

For details and mathematical explanation of the new numeric scale, please see reference [1].

Section 3: Why This Matters (Universality)

From Math -->Physics --> Infrastructure:

* The new scale naturally encodes gauge invariance
* Physics laws reformulated on this scale have universality
* Works at any scale; no preferred reference frame
* This is what physics actually describes - not absolute reality, but relationships.

Universality:

The reformulated physics means:

* The same equations describe phenomena at quantum, human, and cosmic scales.
* Laws that work in any reference frame without modification.

The Key Principles of New Physics

* *Neutral Reference Systems:*

Changes can only be described relative to something else. Neutrality is the idea that there are only constant (small) changes (e.g., a series of proportional grows in nature). For example, when representing momentum, the variable is the rate of change of

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motion. In this case, the neutrality condition translates to uniform acceleration.

Uniform accelerated systems can be viewed as 'being at rest' with respect to a uniformly accelerated reference frame. This concept was originally introduced by Aristotle [6], but is somehow obscured in current physics. The most common example of a neutral reference system is Einstein's free-falling elevator.The free-fall condition means uniform acceleration under local gravitational influences: all objects inside a free-falling elevator are 'at rest' with respect to the elevator, but in a uniformly accelerated motion with respect to a reference system on the Earth.

Describing non-uniform changes

Large variations in both absolute and relative changes are often associated to systems that are far away from equilibrium. The time series of observations showing these changes are stochastic in nature. The measuring variable is a statistical distribution of the expected energy levels - equilibrium states relative to a neutral reference system. Statistical estimates alone cannot be used to draw inferences; they are not conclusive unless validated.

Using the new scale, the principle of entropy maximization can be used to determine an unbiased estimator of the statistical distribution for a neutral reference.

The Law of Changes (Universality)

The action-reaction principle reflects the basic idea that the larger the input *x*, the greater the impact *y* is expected to be.

dy represents the amount of energy exchanged during an interaction. To make comparisons between different magnitudes of changes in *y* more accurately, we can employ the new scale.

The new scale, where numbers represents proportions, allows us to formulate this proportionality relation universally.

The Universal Law of Changes can be formulated as follows:

$$logr\ dy = k*dz$$

where:

k = constant

dz = normalized changes compared against a threshold

r = the smallest noticeable change (the base of the logarithm), which that can be geometrically viewed as displacements on the new scale, where numbers represent proportions.

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Relativity

A notion of “relativity” exists: for two systems (belonging to the same class) having different relative units r1 and r2 , the changes are related by a scaling similarity factor r1 /r2 . This model of physical space was introduced as a standard model for measurable effects of an environmental factors [1]. In this context, concept of relativity means that all “similar” systems are affected in a similar way. The new scale can also be used as an inferential predictive tool. It allows forecasting future changes in system based on a statistical analysis of the history of interactions.

Applications: Electrical Vibrations

An electrical circuit is known to behave similarly to a spring with a mass attached. A spring with a mass attached can vibrate at a natural frequency. Similarly, an electrical circuit with inductance (L) and capacity (C) can oscillate at a resonant frequency (=1/2pi *sqrt(L*C)) .

The electrical capacity can vary, as Tesla explained [6],

“*Since many years scientific men engaged in the study of physics and electrical research have taken it for granted that certain quantities, entering continuously in their estimates and calculations, are fixed and unalterable. The exact determination of these quantities being of particular importance in electrical vibrations.”*

New Physics principles can help measure such changes,

Section 4: Wireless Energy Networks as Expression of True Physics

The Problem with Current Systems:

Current infrastructure which is built on classical mechanics that mirrors the number line: absolute coordinates, fixed origins—fixed power plants, hardwired cable routes, centralized control points.

This creates inherent limitations: brittleness, inefficiency, geographic constraints, and centralized authority.

A Different Approach:

What if infrastructure operated on relational principles instead? Not send power from Point A to Point B, but energy flows through a network of equivalent nodes based on relationships between them - a wireless system that is built on cyclic/proportional logic

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would be distributed, resilient, egalitarian.

This is the logic of:

* Packet-switched energy following cyclic group transformations. Packet-switched networks (the internet(works this way with data.
* Distributed systems: No absolute positions; only relational flows (no single point of failure).
* Self-healing, universally applicable. Scale-invariant design (works equally well in small regions or globally).
* Every node equivalent; no need of central authority.

The Practical Vision: Wireless Energy Networks

* Imagine a global network of relay stations that:
* Transmit energy in directed packets (not broadcast)
* Route packets intelligently around obstacles and failures
* Work equally well over oceans, deserts, remote regions
* Require minimal ground infrastructure compared to cables
* Self-heal when nodes fail (reroute automatically)

Each relay station is functionally equivalent—no hierarchy, no central authority needed. Energy finds its way based on network topology, not fixed routes.

Using ionosphere's conductive properties for wireless energy transmission.

Why Ionosphere?

The ionosphere extends roughly from 80 km-1000 km above the Earth's surface.

In the ionosphere, ultraviolet radiation and X-rays from the Sun strip the electrons from atmospheric gases (ionized atoms and neutral particles primarily oxigen and nitrogen), creating a 'sea of free electrons' and positive ions. Because of these free electrons, ionosphere has 'electrical conductivity' similar to a metal.

This characteristic of ionosphere has been confirmed by:

* Radio signals bounce between Earth's surface and the ionosphere, allows long-distance communication.
* The free electrons in the ionosphere respond to magnetic forces. They interact with the Earth's magnetic field (observed phenomenona: aurora borealis) and protects us from solar radiation.

The ionosphere is essentially a plasma state (sometimes called the 4th state of matter)

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because it is different than gas, liquid and solid.

The ionosphere might be used intelligently to create future wireless energy transmission. Also for monitoring outer space radiation.

Why This Matters Now:

* 71% of Earth is ocean: Wireless solves the submarine cable problem.
* Renewable energy explosion: Need flexible distribution, not rigid grids.
* Developing regions: Can leapfrog traditional infrastructure (like they did with cellular)
* Resilience: Distributed networks survive what centralized systems cannot.

The Path Forward:

Relay stations transmitting power over kilometers, optimizing efficiency, establishing routing protocols.

This isn't distant future—the technology exists. What's needed is reimagining infrastructure as a distributed network rather than a centralized system.

History of the vision of wireless energy

Nikola Tesla's Wardenclyffe Tower (1901-1917) project - one of the most fascinating "what-if" scenarios in the history of technology.

Tesla designed and began constructing a massive tower on Long Island, New York with the goal of transmitting electrical energy wirelessly across the globe. His concept was remarkably ambitious:

* The tower would transmit high-frequency electrical energy upward into the ionosphere
* The energy would propagate around Earth through the ionosphere (which he understood had conductive properties)
* Receiving towers at various locations would capture and convert this energy back into usable electricity

Tesla believed that ionosphere could act as a natural conductor for electricity, similar to how it reflects radio waves. In theory, Earth's surface and the ionosphere would form a giant "electrical circuit."

Tesla's Approach

Tesla envisioned using extremely high voltages and frequencies to excite the ionosphere into a conductive state. He theorized that once this "wireless transmission system" was activated, electrical energy could be beamed and collected anywhere on

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Earth with appropriately tuned receiving equipment.

While Tesla's intuition about the ionosphere's conductive properties is correct, his wireless power transmission scheme faces some challenges:

* Electrical energy radiating into the ionosphere would spread out in all directions (inverse square law), losing intensity rapidly with distance. Unlike focused radio waves (which carry information, not significant power), transmitting usable amounts of electrical energy would require enormous power, with most of it wasted as heat in the atmosphere.
* Could create massive electromagnetic interference, disrupting radio communications, electronics, and potentially biological systems.

Modern Satellite Communication replaced old Ionospheric Reflection

In ionospheric reflection, radio signals bounce off the upper atmosphere to reach distant locations. With satellite communication, signals travel directly upward to an orbiting satellite, which then relays them back down to another location on Earth. This eliminates the need to bounce signals off the ionosphere. The basic process is:

* Uplink: a ground station transmits a radio signal upward to a satellite in orbit.
* Relay: The satellite receives the signal, amplifies it, and changes its frequency (to avoid interference).
* Downlink: The satellite transmits the signal back down to another ground station or directly to receivers on Earth.

This happens almost instantaneously, allowing real-time communication across the globe. This discovery revolutionized current global communication.

Section 5: Beyond Mathematics and Physics to Human experience:

* *The nature of the Universe is essentially energy.*

Substance means that which underlies all outward manifestations, the essence, the essential reality, which is the concept of energy.

As Aristotle explains in [8] “an attribute of a substance that belongs to the subject should not be identified to be anything but of one of many possible characteristics of particular substance”. He further adds that, “those who understand something having an attribute as being something” are wrong. “If this would be so, there is nothing to prevent these things [substance] as being many things since” substance is plainly divisible into other substances.”

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* *The new physics describes energy or force.*

Not force as the materialists use the terms, for their concept of force has a vector representation on traditional number line.

* *Material Universe, all phenomena that are apparent to our material senses is subjective - relative to the subject (observer).* While the Universe as a whole is infinite, our conceptual world is a measurable, a subjective mental model. This principle establishes the subjective nature of our conceptual world.
* *"Everything changes. Nothing is at rest"* (Heraclitus).

Reference - the neutral observer.

* *Cyclical nature of things. Everything flows, in and out. Everything material is relative. The difference is merely in degree.*

This aligns with the observation that everything vibrates. The degrees of the rate of vibrations constitute the degrees of measurement on the new cyclical scale- the scale of vibrations, where quantitative characteristic of a type is its vibrational signature - in other words, the degrees of the 4th dimension.

* *Another Dimension.* The absolute (space-time) and relative scales are complementary, and the new and physics coexist. However, the new scale adds dimension that changes the model of the world.

'Godly Freedom Through Alignment with Reality'

* Current systems constrain us because they're built on false fundamental principles.
* When infrastructure mirrors what physics actually reveals, freedom emerges naturally.
* Abundance without scarcity- human flourishing through abundance and autonomy.
* Autonomy without chaos (because relationships are preserved).
* Freedom as a natural consequence of aligning with how reality actually works.
* Spiritual/transcendent freedom (liberation from artificial constraints)

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